

Eclipse Soundscapes Data Management: Data Processing (Stage 2)

A methods and provenance reference for people using Eclipse Soundscapes (ES) datasets

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1. What This Document Covers

This document describes Stage 2 (Data Processing) of the Eclipse Soundscapes (ES) data lifecycle. It explains how audio data returned on participant-submitted microSD cards were ingested, verified, validated, and organized after physical receipt and sorting in Stage 1. Stage 2 included device-to-site verification, timestamp validation, human-in-the-loop review of incomplete or invalid timing information, metadata reconciliation, and preparation of datasets for downstream analysis and public archiving workflows.

Eclipse Soundscapes is a NASA-funded participatory science project that deployed AudioMoth acoustic recorders during two eclipses: the October 14, 2023 annular solar eclipse and the April 8, 2024 total solar eclipse. Volunteers recorded environmental soundscapes across the United States so that researchers could study how rapid eclipse-related light changes affected animal behavior and ambient sound.

Stage 2 served as the primary validation and processing stage within the ES data lifecycle. Building upon the site-level records assembled during Stage 1, this stage examined the contents of returned audio datasets, verified the relationship between recordings and ES ID # site records, evaluated timestamp plausibility relative to eclipse timing windows, documented participant-provided timing information when available, and identified datasets suitable for downstream scientific analysis. Throughout this process, all raw audio data was preserved without modification.

Specifically, this document details:

- audio data ingestion and device-to-site verification
- timestamp validation and eclipse-window plausibility checks
- human-in-the-loop review procedures
- metadata reconciliation and timing documentation
- provenance, integrity, and backup procedures
- preparation of datasets for Stage 3 public archiving and Stage 4 scientific analysis

This document is intended as a methods and provenance reference for researchers, archivists, developers, and future project teams seeking to understand how large-scale participatory science audio datasets were validated, documented, and preserved while maintaining clear separation between raw data and interpretive metadata.

2. What Stage 2 Produced

Stage 2 transformed organized site-level records from Stage 1 into validated and documented audio datasets suitable for long-term preservation, downstream analysis, and public archiving.

- **Validated Raw Audio Datasets Organized by ES ID #**
Raw WAV audio recordings preserved without modification and organized into structured site-level datasets linked to ES ID # records.
- **Expanded Master Data Collectors Spreadsheet**
The Master Data Collectors Spreadsheet was further expanded during Stage 2 to incorporate timestamp validation outcomes, participant-provided timing information, device verification results, and processing notes associated with each recording site.
- **Analysis-Eligible Timestamped Dataset Subset**
A subset of datasets identified as containing valid AudioMoth-generated timestamps suitable for downstream Stage 4 scientific analysis workflows.

- **Open-Source Audio Processing and Validation Software**
Development and operational deployment of the Eclipse Soundscapes WAV Audio Validation & Extraction Suite (ES WAVES) and the Eclipse Soundscapes AudioMoth Metadata Extractor Suite (ES AMES), which supported scalable ingestion, metadata extraction, timestamp validation, and organization of large-scale AudioMoth datasets.
- **Structured Backup and Integrity Verification Records**
SHA-512 integrity hashes and structured backup copies supporting long-term preservation and verification of processed datasets.

These outputs preserved the original audio data while documenting validation outcomes and preparing datasets for both public archiving in Stage 3 and scientific analysis workflows in Stage 4.

3. Why Stage 2 Was Needed

Stage 2 was necessary because the organized site-level records created during Stage 1 did not yet verify the contents, usability, or timestamp validity of the returned audio data. Although Stage 1 established the relationship between participant submissions and recording sites, the audio files themselves had not yet been systematically examined to confirm audio data-to-site linkage, verify embedded AudioMoth timestamps, evaluate audio usability, or determine which datasets were suitable for downstream scientific analysis.

Because AudioMoth devices embed timestamps directly in WAV file names and metadata, the scientific usability of each dataset depended on whether those timestamps were recorded correctly by the device. In practice, returned datasets included a range of conditions, including valid timestamps, missing timestamps, unreadable files, and clearly incorrect timestamps resulting from device clock failures.

Stage 2, therefore, introduced centralized ingestion, automated validation, metadata extraction, timestamp evaluation, and human-in-the-loop review workflows. These processes enabled the project to preserve the original audio data while documenting timestamp conditions, incorporating participant-provided timing information when available, and identifying datasets appropriate for downstream analysis workflows.

This stage also addressed the practical challenge of efficiently and consistently processing large volumes of returned microSD cards and WAV files. To support scalable processing, the project developed centralized ingestion infrastructure along with the Eclipse Soundscapes WAV Audio Validation & Extraction Suite (ES WAVES) and the Eclipse Soundscapes AudioMoth Metadata Extractor Suite (ES AMES).

Stage 2 established the validated, documented, and organized audio dataset structure required for long-term preservation, public archiving, and scientific analysis.

The workflow diagram below provides a Stage 2–focused view of the Eclipse Soundscapes data lifecycle, showing only the directly relevant deliverables from related stages. The complete ES Operational Infrastructure and Data Stewardship Workflows diagram is included in [Appendix 1](#).

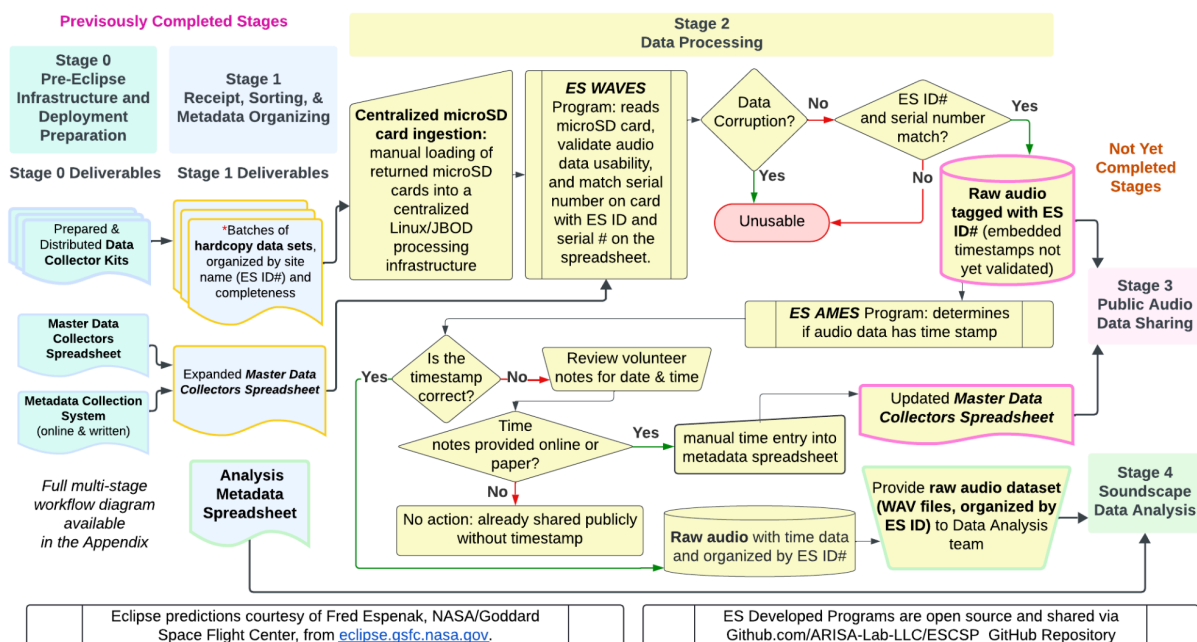


Figure 1. Eclipse Soundscapes Stage 2 workflow for validating, processing, and organizing participant-submitted audio datasets. The workflow illustrates how returned microSD card data were ingested into the centralized processing infrastructure, validated using the ES WAVES and ES AMES software tools, reviewed for timestamp integrity and device verification, and organized into structured ES ID# datasets for downstream public archiving and scientific analysis.

4. Infrastructure for Centralized Processing

To support the scalable processing required for Stage 2, the team developed a centralized data ingestion system capable of reading and processing multiple microSD cards simultaneously. This system consisted of a low-cost, low-power Linux server with direct-attached JBOD (Just a Bunch of Disks) storage connected to two 10-slot USB hubs, allowing up to 20 microSD cards to be read simultaneously.

Audio data from these cards was processed using the ES WAV Audio Validation & Extraction Suite (ES WAVES), an internally developed software tool available through the Eclipse Soundscapes GitHub repository (<https://github.com/ARISA-Lab-LLC/ESCSP-ES-WAV-Audio-Validation-Extraction-Suite>). This system enabled automated validation and extraction of audio file information across multiple devices in parallel.

This approach replaced an earlier workflow that processed one card at a time, which was not feasible for scalable processing. This centralized system enabled efficient ingestion and validation of audio data within the Stage 2 pipeline.

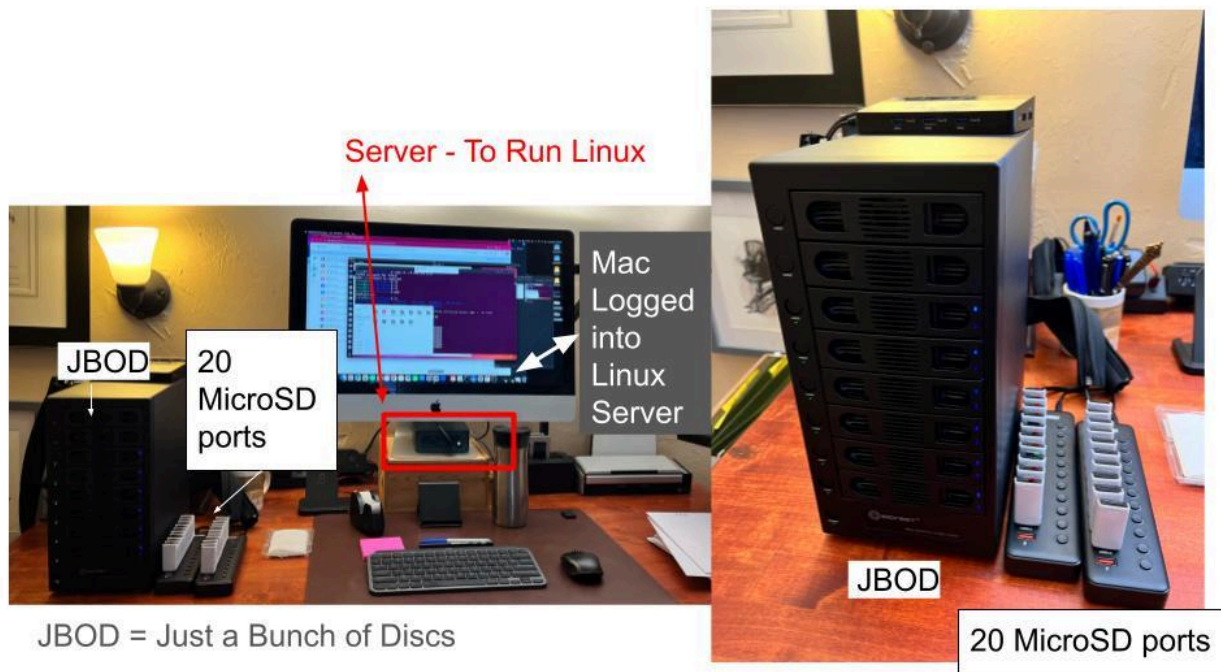


Figure 2. Data ingestion concept: a Linux server accessed via an iMac, Just a Bunch of Discs (JBOD) array storage, and multi-card microSD USB hubs allowing for automated software to process 20 microSD cards at a time.

5. Process Design Decisions and Scalability Lessons

The design of the Stage 2 processing workflow was informed by prior implementation during the 2023 annular eclipse data collection. Several process considerations were incorporated to support efficient and consistent handling of returned audio data.

These included reducing operator input through automation (for example, using software to name and sort audio files rather than manual labeling), batch processing of data rather than single-card workflows, and separating tasks requiring human judgment into discrete review queues. The use of batch processing, introduced in Stage 1 for organizing returned microSD cards, was extended in Stage 2 to support scalable ingestion and validation of audio data.

In addition, the processing workflow was designed to operate entirely on internally managed systems, including local ingestion hardware, direct-attached JBOD storage, and the ES WAV Audio Validation & Extraction Suite (ES WAVES), rather than relying on external platforms or services.

6. Inputs to Stage 2

Stage 2 begins after Stage 1 has physically sorted returned microSD contents into site-level working areas organized by ES ID #. The key inputs are:

Raw WAV recordings: Uncompressed audio files recorded by AudioMoth devices are stored in WAV format files. These files contain the acoustic data used in Stage 2 processing.

AudioMoth Recording Specifications (Context for Stage 2 Inputs)

AudioMoth devices used in the Eclipse Soundscapes project were used in “Default” mode, which is configured to record uncompressed audio in WAV format at a sampling rate of 44.1 kHz (capturing frequencies from 0 to 22.05 kHz, covering the full range of human hearing). At this rate, the device produces approximately 8.59 GB of data per day, or roughly 60 GB over a seven-day recording window. Files were stored on 64 GB SanDisk Extreme microSDXC cards. (**Note:** While the data collection protocol only called for a 5-day recording window, a margin was added for extra data collection days.) For full AudioMoth device specifications, including firmware details and recording configurations, visit <https://www.openacousticdevices.info/audiomoth>

CONFIG.TXT files: Each AudioMoth automatically generates a *CONFIG.TXT* file at the time of recording. This file contains the device serial number, recording settings, temperature, and battery voltage information. This file is the primary mechanism for programmatically linking a returned microSD card back to its registered ES ID #, because device serial numbers were recorded in the project registration *Master Data Collectors Spreadsheet* before kits were mailed to participants. If there was an error in correlating a device serial number to values in the *Master Data Collectors Spreadsheet*, an error was flagged by *ES WAVES* so that a human could provide a visual inspection of the return materials in conjunction with the *Master Data Collectors Spreadsheet*.

Registration metadata: The *Master Data Collectors Spreadsheets* maintained during Stages 1 and 2 linked each ES ID # to its device serial number, kit assignment, geographic coordinates, and eclipse timing parameters. The versions of these spreadsheets that contain participant names and contact information are not publicly shared.

Participant-provided notes and survey fields: When device timestamps were incomplete or missing (for example, due to time-chime failures), participant-submitted survey responses and handwritten notes provided the primary evidence for documenting recording start times. Participants were instructed to record the exact time they turned on their AudioMoth as a fallback.

For full AudioMoth device specifications, including firmware details and recording configurations, see the separate Eclipse Soundscapes AudioMoth Specifications reference document. The summary above provides enough context for understanding Stage 2 processing.

7. Processing Steps

Stage 2 processing began with the ingestion of audio data via returned microSD cards and proceeded through validation and organization of audio data and time-related metadata. The subsections below walk through the workflow used to validate audio data usability, confirm audio data-to-site linkage, evaluate timestamp information, document participant-provided timing data where available, and prepare datasets for downstream analysis.

7.1 Data Ingestion, Audio Validation, and Serial Number Verification

Returned microSD cards were ingested into the centralized processing system by inserting them into multi-card USB hubs connected to a low-cost server with a JBOD storage system. Audio data were then processed using the ES WAV Audio Validation & Extraction Suite (ES WAVES).

ES WAVES read the contents of each microSD card, including WAV audio files and CONFIG.TXT files, and performed initial validation of audio data usability. This included verifying that audio files were present, readable, and consistent with expected recording formats.

As part of this process, ES WAVES extracted device serial numbers from the CONFIG.TXT files and compared them to the corresponding serial numbers and ES ID # values recorded in the Master Data Collectors Spreadsheet. This step established the initial linkage between audio data and site-level records.

Records with missing, unreadable, or inconsistent audio data, as well as mismatches between device serial numbers and spreadsheet records, were flagged as unusable.

7.2 Timestamp Validation

Recording timestamps embedded in AudioMoth-generated WAV file names (formatted as YYYYMMDD_hhmmss.WAV) were evaluated for completeness and plausibility relative to the expected eclipse window at each recording location. A “plausible” timestamp falls within the expected date range near the eclipse window across files from the same site. These timestamps represent the start time of each audio recording as recorded by the device’s internal clock.

The ES AudioMoth Metadata Extractor Suite (ES AMES) was used to determine whether audio files contained valid timestamp information based on file naming conventions and extracted metadata

Timestamps were reviewed to determine whether:

- a timestamp was present
- the timestamp fell within a plausible temporal range for the recorded eclipse event
- the timestamp contained clearly incorrect values (e.g., incorrect year)

A common AudioMoth failure mode produces timestamps that are decades away from the eclipse window (for example, defaulting to epoch-era values near January 1, 1970, when the time chime procedure fails to set the device clock). These anomalous timestamps were automatically flagged for manual review.

Timestamp plausibility validation also checked alignment with the eclipse timing window, using timing materials from NASA’s Goddard Space Flight Center (<http://eclipse.gsfc.nasa.gov/>).

Where timestamps were missing or clearly incorrect, their condition was documented, and any participant-reported recording time information available from online forms or handwritten notes was added to the Master Data Collectors Spreadsheet. No modifications were made to the timestamps embedded in the audio files. This preserved the original recordings while providing accompanying timing information that could support future interpretation.

7.3 Human-In-The-Loop Review

Records flagged during timestamp validation were reviewed manually to document available information about recording start times.

When automated timestamp validation failed, Stage 2 used a hierarchy of evidence to document what was known about the recording timeline:

Primary evidence: Survey fields submitted by the participant, including recording start time notes entered through an online survey form available via a link on the Data Collector page on the Eclipse Soundscapes website.

Secondary evidence: Handwritten notes included with returned microSD cards or kits, often containing start/end recording times, latitude and longitude, and other contextual information.

No reliable evidence: If neither survey fields nor participant notes provided usable time information, the record was documented as having incomplete timestamp data. The audio data were still shared publicly, but the missing timestamp information was noted in the metadata record.

Manual review outcomes were explicitly documented in the *Master Data Collector Spreadsheet* so that timing data will be available either through device-native timestamps or via shared participant-provided evidence. Participant shared recording start dates and times were incorporated into the *Master Data Collectors Spreadsheet*.

8. Data Integrity: No Modification of Audio Files

Key provenance statement for researchers: The WAV audio files shared in the public archive are byte-for-byte identical to the contents of the returned microSD cards. No timestamp corrections, format conversions, or other modifications were applied to the shared audio data.

Stage 2 processing did not alter the audio recordings. All validation, review, and documentation steps were performed without modifying the original files or their embedded metadata. The only changes introduced during Stage 2 were in the form of metadata documentation. Recording time information, when available, was documented in the Master Data Collectors Spreadsheet based on device-generated timestamps and participant-provided notes. Where timestamp information was missing or clearly incorrect, this condition was recorded, but the original data were left unchanged.

9. Data Integrity and Backups

Stage 2 created structured backup copies of verified data before packaging for long-term archiving. The project maintained three backup copies across at least two different types of storage media, with at least one copy at a separate physical location (the 3-2-1 backup strategy).

For each backup file, integrity was verified using SHA-512 hashes that were computed for each site-level audio file. SHA-512 was chosen over faster legacy hash algorithms (such as MD5) because it provides stronger collision resistance and is better suited for long-term public archiving workflows where integrity verification is vital.

10. Stage 4 Analysis Eligibility

All data processed in Stage 2 was carried forward to Stage 3 for public sharing. No datasets were excluded from the public archive based on completeness or quality. For downstream scientific analysis (Stage 4), a subset of the data was provided, consisting only of audio recordings with valid device-generated timestamps. This subset was identified during Stage 2 using the ES AudioMoth Metadata Extractor Suite (ES AMES), which was used to determine the presence of usable timestamp information in the audio files.

Of the 575 DOI-assigned site-level datasets published on Zenodo, 353 verified sites met all three criteria. These datasets, identified based on the presence of valid device-generated timestamps, formed the subset shared with the analysis team. The remaining sites are publicly available and fully documented via Stage 3.

11. Common Issues Encountered

The following issues were encountered during Stage 2 processing. In every case, the data were carried forward to Stage 3 for public sharing. This section is provided so that researchers working with ES datasets can interpret metadata flags and understand the provenance of specific records.

Issue	How it was handled
Missing or unmountable microSD card	Recorded as missing. Routed for follow-up where possible. Data could not be recovered from unmountable cards, but the site record was preserved with available registration metadata.
Serial number mismatch or not found	Routed to manual review. Resolved when other evidence was consistent (e.g., <i>CONFIG.TXT</i> and <i>AME</i> data matched but spreadsheet had a transcription error). Documented as unresolved when attribution could not be validated.
MicroSD card renamed (e.g., "NO NAME")	Automation could not locate expected file paths on renamed cards. Records were queued for manual handling and the card contents were processed once the correct paths were identified.
Time-chime failure or clock unset	Timestamps defaulted to epoch-era values (e.g., 1970). Participant survey fields and notes were consulted to document the actual recording start time. If no reliable evidence existed, no timestamp state was recorded.
0-byte or corrupted files	Flagged during processing. Usable portions of the site record were retained if the remaining files were consistent and attributable.
Device failure during eclipse	Site data were carried forward to Stage 3 but not shared, as the subset recordings were carried forward to Stage 4.

Table 2. Common issues encountered during Stage 2 processing and how they were handled.

12. Glossary

This glossary is shared across the Eclipse Soundscapes Stage 0–3 data management documents and defines terminology used throughout the ES data lifecycle workflow.

Term	Definition
AME / ES AMES	ES AudioMoth Metadata Extractor Suite. A software tool used to extract and evaluate embedded AudioMoth metadata, including timestamps, serial numbers, firmware information, and recording details to support validation and provenance workflows.
annular solar eclipse	when the Moon blocks the center of the Sun from view. The Moon appears to be too small to completely block the Sun.
API	Application Programming Interface. A structured method that allows software systems to communicate programmatically with external platforms or services.
Arbimon	An acoustic data platform originally considered by Eclipse Soundscapes for data storage, browsing, and analysis prior to the project's transition to Zenodo.
AudioMoth	An open-source, low-cost acoustic recording device used by ES volunteers to capture environmental soundscapes during eclipse events.
AZUS	Automated Zenodo Upload Software. A configuration-driven, open-source software pipeline developed by ARISA Lab to prepare, package, document, validate, and upload Eclipse Soundscapes datasets to Zenodo.
batch processing	A workflow approach that processes large groups of datasets simultaneously rather than handling one dataset at a time.
CONFIG.TXT	An AudioMoth-generated configuration file containing device settings such as sample rate, gain, firmware version, serial number, recording schedule, temperature, and battery state.
CSV	Comma-Separated Values. A simple, human- and machine-readable text file format used for spreadsheets and structured tabular data.
data dictionary	A structured reference file defining variable names, descriptions, data types, units, sources, code meanings, and notes associated with a dataset or metadata file.
dataset DOI / data site DOI	A persistent Digital Object Identifier assigned to a published Zenodo dataset record, allowing the dataset to be cited, accessed, and reused over time.

dry-run mode	A validation mode used by AZUS to verify configuration settings, metadata, credentials, and file paths before live uploads occur.
eclipse contact times	The calculated timing of eclipse phases for a specific site, including first contact, second contact, maximum eclipse, third contact, and fourth contact.
eclipse maximum	The moment of greatest solar obscuration at a specific location during a solar eclipse.
eclipse percent	The percentage of solar obscuration at a specific location.
ES ID #	Eclipse Soundscapes Identification Number. A unique numeric identifier assigned to each recording site, allowing datasets to be publicly shared without exposing participant identity.
ES WAVES	Eclipse Soundscapes WAV Audio Validation & Extraction Suite. A software system that ingests audio data from multiple microSD cards simultaneously, validates WAV files, extracts metadata, organizes datasets, and supports downstream analysis workflows.
FAIR principles	A framework for scientific data stewardship emphasizing that datasets should be Findable, Accessible, Interoperable, and Reusable.
file_list.csv	A manifest file included in each Zenodo record that lists every file, file size, file type, description, SHA-512 hash, and associated data dictionary.
first contact	solar eclipse Start (First Contact). The time at which the Moon first begins to obscure the Sun.
fourth contact	Eclipse End (Fourth Contact) The time at which the Moon completely stops blocking the Sun from view.
Hash / SHA-512 Hash	A cryptographic fingerprint used to verify file integrity. ES used SHA-512 hashes to confirm that files were not altered or corrupted during storage, transfer, or download.
Human-In-The-Loop review	A workflow step requiring manual human review and interpretation when automated validation systems identify incomplete, inconsistent, or ambiguous data conditions.
HTML README template	A reusable template used to generate standardized, site-specific Zenodo record descriptions and README documentation through automated metadata substitution.

InvenioRDM	The open-source digital repository framework underlying Zenodo. AZUS communicates with Zenodo through the InvenioRDM REST API.
JBOD	“Just a Bunch of Disks.” A direct-attached storage system consisting of multiple hard drives used for scalable local data storage.
JSON configuration file	A human-readable configuration file used by AZUS to define project metadata, upload settings, templates, required metadata fields, file paths, and workflow behavior.
Level 0 data	Raw, unprocessed data preserved in its original form without timestamp correction, resampling, filtering, or modification.
metadata	Structured descriptive information about data, including location, timestamps, eclipse parameters, device information, and processing documentation.
metadata reconciliation	The process of comparing and integrating metadata from multiple sources to resolve inconsistencies and establish validated site-level records.
microSD Card	A removable flash memory storage device used within AudioMoth recorders to store WAV audio recordings and configuration files.
participatory science	Scientific research conducted with participation from members of the public contributing observations, data collection, or analysis activities.
PII	Personally Identifiable Information. Protected personal information that could uniquely identify an individual participant.
provenance	Documentation describing the origin, processing history, validation status, and stewardship history of a dataset.
README	Human-readable documentation accompanying a dataset that explains its contents, structure, collection methods, metadata, and usage guidelines.
related identifiers	Structured citation metadata linking Zenodo records to related datasets, publications, software repositories, reports, or videos.
REST API	A web-based application programming interface that allows software systems to communicate using standardized HTTP requests.
scalability	The ability of a workflow, infrastructure system, or software tool to efficiently support increasing amounts of data, users, or processing demand.
second contact	Totality Start (for sites within the path of totality). The time at which totality begins during a total solar eclipse.

site-level dataset	A complete dataset associated with a single Eclipse Soundscapes recording location identified by one ES ID #.
staging directory	A temporary preparation directory created during AZUS processing that contains the ZIP archive and all companion files ready for upload to Zenodo.
subject labels	Standardized Zenodo metadata tags used to categorize datasets by eclipse event, analysis relevance, site type, or project designation.
template substitution	An automated process in which variable placeholders within templates are replaced with site-specific metadata values during dataset generation.

13. References and Related Resources

These references and related resources are shared across the Eclipse Soundscapes (ES) Stage 0–3 data management documents and provide supporting literature, technical resources, software references, infrastructure documentation, and related materials referenced throughout the ES data lifecycle workflow.

Core Eclipse Soundscapes Resources

Eclipse Soundscapes Zenodo Community

Public archive of Eclipse Soundscapes datasets, documentation, and related resources.

<https://zenodo.org/communities/eclipsesoundscapes>

ARISA Lab Eclipse Soundscapes GitHub Repository

Open-source software, workflows, processing tools, and supporting documentation for the Eclipse Soundscapes project. <https://github.com/ARISA-Lab-LLC/ESCSP>

Eclipse Soundscapes Technical Workflow References

Severino, M., & Winter, H. (2026). Eclipse Soundscapes Data Collector Role Training and Implementation Manual (2023–2024). Zenodo. <https://doi.org/10.5281/zenodo.18623443>

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Winter, H. T., & Severino, M. Eclipse Soundscapes Data Management: Public Data Sharing (Stage 3). Zenodo. <https://doi.org/10.5281/zenodo.18683437>

Eclipse Timing and Astronomical Resources

NASA Goddard Space Flight Center Eclipse Web Site

Eclipse timing predictions and eclipse path calculations courtesy of Fred Espenak, NASA/GSFC. <https://eclipse.gsfc.nasa.gov/eclipse.html>

Wheeler, W. M., et al. (1935). Animal Behavior During a Total Solar Eclipse. Boston Society of Natural History.

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AudioMoth Operation Manual. (2022). Open Acoustic Devices.

theteam@openacousticdevices.info. Retrieved from:

<https://www.openacousticdevices.info/applications>

Open Acoustic Devices (AudioMoth)

AudioMoth device documentation, firmware information, and technical specifications.

<https://www.openacousticdevices.info/>

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Soundscape Ecology and Acoustic Science References

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Open Science and Repository Infrastructure Resources

Zenodo

General-purpose open-access repository operated by CERN and OpenAIRE for long-term data preservation and DOI assignment. <https://zenodo.org>

FAIR Principles

Findable, Accessible, Interoperable, and Reusable data stewardship principles.

<https://www.go-fair.org/fair-principles/>

InvenioRDM

Open-source repository framework underlying Zenodo and used by AZUS through the REST API. <https://inveniordm.docs.cern.ch/>

Zenodo REST API Documentation

Documentation for Zenodo's Open API used by AZUS for automated dataset publication workflows.

<https://developers.zenodo.org/>

Arslan, R. C. (2019). How to Automatically Document Data With the Codebook Package to Facilitate Data Reuse. *Advances in Methods and Practices in Psychological Science*, 2(2), 169–187. <https://doi.org/10.1177/2515245919838783>

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<https://www.ooma.com/blog/average-us-internet-speeds-over-time/>

Appendix 1: ES Operational Infrastructure and Data Stewardship Workflows

